

Pilot-Based Power Analysis

Pilot Study Team

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```
library(data.table)
library(sandwich)
library(lmtest)
library(ggplot2)
library(knitr)
```

1. Read Pilot Data

We start from the processed pilot dataset created by `process_pilot_csv.R`.

```
PILOT_FILE <- "pilot_data.csv"
pilot <- fread(PILOT_FILE)

# Sanitize Unicode for PDF/LaTeX (e.g. >= from survey text like ">= 3 times per week")
sanitize_ascii <- function(x) {
  if (!is.character(x)) return(x)
  x <- gsub("\u2265", ">=", x, fixed = TRUE) # >= (greater-or-equal)
  x <- gsub("\u2264", "<=", x, fixed = TRUE) # <= (less-or-equal)
  x <- gsub("\u2013", "-", x, fixed = TRUE) # en-dash
  x <- gsub("\u2014", "-", x, fixed = TRUE) # em-dash
  x
}
for (j in seq_along(pilot)) if (is.character(pilot[[j]])) set(pilot, j = j, value = sanitize_ascii(pilot[[j]]))
str(pilot)

## Classes 'data.table' and 'data.frame':  8 obs. of  38 variables:
## $ email
## $ reg_Timestamp
## $ reg_Age
## $ reg_Do.you.have.any.injuries.or.medical.conditions.that.would.make.performing.a.plank.exercise.un
## $ reg_Are.you.available.to.complete.two.sessions..each.lasting.no.more.than.10.minutes..at.least.24
## $ reg_I.have.read.the.study.description.and.agree.to.participate.
## $ reg_Experiment_ID
## $ reg_Column.1
## $ experiment_id
## $ s1_Timestamp
## $ s1_What.is.your.age.
## $ s1_What.is.your.gender.
## $ s1_Are.you.a.regular.gym.goer.or.physically.active.
## $ s1_What.type.of.physical.activity.do.you.usually.do.
## $ s1_Do.you.regularly.perform.planks.
## $ s1_How.long.were.you.able.to.hold.your.plank..in.seconds...
```

```
## $ s1_After.the.music.began..but.before.you.started.the.plank..how.physically.energized.did.you.feel
## $ s1_After.the.music.began..but.before.you.started.the.plank..how.motivated.were.you.to.perform.wel
## $ s1_Immediately.after.completing.the.plank..how.hard.did.the.exercise.feel.overall...0...no.exerti
## $ s1_Did.you.use.headphones.during.the.plank.exercise.
## $ s1_Did.you.pause.or.restart.the.plank.at.any.point.
## $ s1_If.yes..please.briefly.describe.what.happened..
## $ s1_Were.the.music.volume.instructions.clear.
## $ tte_session1
## $ s2_Timestamp
## $ s2_How.long.were.you.able.to.hold.your.plank..in.seconds...
## $ s2_After.the.music.began..but.before.you.started.the.plank..how.physically.energized.did.you.feel
## $ s2_After.the.music.began..but.before.you.started.the.plank..how.motivated.were.you.to.perform.wel
## $ s2_Immediately.after.completing.the.plank..how.hard.did.the.exercise.feel.overall...0...no.exerti
## $ s2_Did.you.use.headphones.during.the.plank.exercise.
## $ s2_Did.you.pause.or.restart.the.plank.at.any.point.
## $ s2_If.yes..please.briefly.describe.what.happened..
## $ s2_How.easy.was.it.for.you.to.follow.the.instructions.provided.in.the.PDF.document...1...Very.dif
## $ s2_How.would.you.rate.your.overall.experience.participating.in.this.study...1...Very.negative..5.
## $ s2_Do.you.have.any.comments.or.suggestions.about.your.experience.in.this.study.
## $ tte_session2
## $ tte_slow
## $ tte_high
## - attr(*, ".internal.selfref")=<externalptr>
```

```
head(pilot)
```

```
##          email      reg_Timestamp reg_Age
##          <char>      <char>      <int>
## 1:      cschrupp@gmail.com 2/28/2026 8:07:46      46
## 2:      devamsheth0806@gmail.com 2/28/2026 9:23:26      26
## 3:      eduardo111081@gmail.com 2/28/2026 8:38:13      44
## 4:      karankpatel192@gmail.com 2/28/2026 7:29:23      26
## 5:      neochrix@yahoo.es 2/28/2026 13:28:44      45
## 6: ryan.castillo256@gmail.com 2/28/2026 18:16:57      29
##      reg_Do.you.have.any.injuries.or.medical.conditions.that.would.make.performing.a.plank.exercise.un
##
## 1:
## 2:
## 3:
## 4:
## 5:
## 6:
##      reg_Are.you.available.to.complete.two.sessions..each.lasting.no.more.than.10.minutes..at.least.24
##
## 1:
## 2:
## 3:
## 4:
## 5:
## 6:
##      reg_I.have.read.the.study.description.and.agree.to.participate.
##                                     <char>
## 1:                                     Yes, I agree
## 2:                                     Yes, I agree
## 3:                                     Yes, I agree
```

```

## 4: Yes, I agree
## 5: Yes, I agree
## 6: Yes, I agree
## reg_Experiment_ID reg_Column.1 experiment_id s1_Timestamp
## <int> <lgcl> <int> <char>
## 1: 2 NA 2 2/28/2026 12:12:55
## 2: 1 NA 1 2/28/2026 10:51:17
## 3: 2 NA 2 3/1/2026 15:59:34
## 4: 1 NA 1 2/28/2026 16:46:28
## 5: 1 NA 1 3/1/2026 14:20:04
## 6: 1 NA 1 3/1/2026 18:10:49
## s1_What.is.your.age. s1_What.is.your.gender.
## <int> <char>
## 1: 46 Male
## 2: 26 Male
## 3: 44 Male
## 4: 26 Male
## 5: 45 Male
## 6: 29 Male
## s1_Are.you.a.regular.gym.goer.or.physically.active.
## <char>
## 1: Yes, regularly (>= 3 times per week)
## 2: No, rarely or never
## 3: Yes, regularly (>= 3 times per week)
## 4: No, rarely or never
## 5: Yes, regularly (>= 3 times per week)
## 6: Yes, regularly (>= 3 times per week)
## s1_What.type.of.physical.activity.do.you.usually.do.
## <char>
## 1: Running, weight training
## 2: Walking and running
## 3: Walk
## 4: Walking, Running
## 5: Yoga
## 6: Volleyball, gym
## s1_Do.you.regularly.perform.planks.
## <char>
## 1: No, rarely or never
## 2: No, rarely or never
## 3: No, rarely or never
## 4: No, rarely or never
## 5: No, rarely or never
## 6: No, rarely or never
## s1_How.long.were.you.able.to.hold.your.plank..in.seconds...
## <int>
## 1: 81
## 2: 76
## 3: 109
## 4: 80
## 5: 160
## 6: 121
## s1_After.the.music.began..but.before.you.started.the.plank..how.physically.energized.did.you.feel
##
## 1:

```

```

## 2:
## 3:
## 4:
## 5:
## 6:
## s1_After.the.music.began..but.before.you.started.the.plank..how.motivated.were.you.to.perform.wel
##
## 1:
## 2:
## 3:
## 4:
## 5:
## 6:
## s1_Immediately.after.completing.the.plank..how.hard.did.the.exercise.feel.overall...0...no.exerti
##
## 1:
## 2:
## 3:
## 4:
## 5:
## 6:
## s1_Did.you.use.headphones.during.the.plank.exercise.
##                                     <char>
## 1:                                     No
## 2:                                     Yes
## 3:                                     No
## 4:                                     Yes
## 5:                                     Yes
## 6:                                     No
## s1_Did.you.pause.or.restart.the.plank.at.any.point.
##                                     <char>
## 1:                                     No
## 2:                                     No
## 3:                                     No
## 4:                                     No
## 5:                                     No
## 6:                                     No
## s1_If.yes..please.briefly.describe.what.happened..
##                                     <lgcl>
## 1:                                     NA
## 2:                                     NA
## 3:                                     NA
## 4:                                     NA
## 5:                                     NA
## 6:                                     NA
## s1_Were.the.music.volume.instructions.clear. tte_session1      s2_Timestamp
##                                     <char>      <int>      <char>
## 1:                                     Yes      81 3/1/2026 17:41:13
## 2:                                     Yes      76 3/1/2026 18:13:08
## 3:                                     Yes      109
## 4:                                     Yes      80 3/1/2026 17:23:12
## 5:                                     Yes      160 3/1/2026 18:14:23
## 6:                                     Yes      121
## s2_How.long.were.you.able.to.hold.your.plank..in.seconds...

```

```

##                                     <int>
## 1:                                     105
## 2:                                     93
## 3:                                     NA
## 4:                                     84
## 5:                                    208
## 6:                                     NA
##   s2_After.the.music.began..but.before.you.started.the.plank..how.physically.energized.did.you.feel
##
## 1:
## 2:
## 3:
## 4:
## 5:
## 6:
##   s2_After.the.music.began..but.before.you.started.the.plank..how.motivated.were.you.to.perform.wel
##
## 1:
## 2:
## 3:
## 4:
## 5:
## 6:
##   s2_Immediately.after.completing.the.plank..how.hard.did.the.exercise.feel.overall...0...no.exerti
##
## 1:
## 2:
## 3:
## 4:
## 5:
## 6:
##   s2_Did.you.use.headphones.during.the.plank.exercise.
##                                     <char>
## 1:                                     No
## 2:                                     Yes
## 3:
## 4:                                     Yes
## 5:                                     Yes
## 6:
##   s2_Did.you.pause.or.restart.the.plank.at.any.point.
##                                     <char>
## 1:                                     No
## 2:                                     No
## 3:
## 4:                                     No
## 5:                                     No
## 6:
##   s2_If.yes..please.briefly.describe.what.happened..
##                                     <lgcl>
## 1:                                     NA
## 2:                                     NA
## 3:                                     NA
## 4:                                     NA
## 5:                                     NA

```

```
## 6: NA
## s2_How.easy.was.it.for.you.to.follow.the.instructions.provided.in.the.PDF.document...1...Very.dif.
##
## 1:
## 2:
## 3:
## 4:
## 5:
## 6:
## s2_How.would.you.rate.your.overall.experience.participating.in.this.study...1...Very.negative..5.
##
## 1:
## 2:
## 3:
## 4:
## 5:
## 6:
## s2_Do.you.have.any.comments.or.suggestions.about.your.experience.in.this.study.
## <char>
## 1: Reorder the instruction pdf to make the forms easier to find
## 2:
## 3:
## 4:
## 5:
## 6:
## tte_session2 tte_slow tte_high
## <int> <int> <int>
## 1: 105 105 81
## 2: 93 76 93
## 3: NA NA 109
## 4: 84 80 84
## 5: 208 160 208
## 6: NA 121 NA
```

2. Estimate Effect Size from Pilot

We use only participants with both slow and high-tempo plank times.

```
# Complete cases for both conditions (needed for within-subjects and for estimating SDs)
pilot_cc <- pilot[!is.na(tte_slow) & !is.na(tte_high)]

if (nrow(pilot_cc) < 4) {
  stop("Need at least 4 complete cases in pilot_data.csv to estimate parameters.")
}

# Between-subjects: marginal means and SDs for slow vs high conditions
mean_slow <- pilot_cc[, mean(tte_slow, na.rm = TRUE)]
mean_high <- pilot_cc[, mean(tte_high, na.rm = TRUE)]
sd_slow <- pilot_cc[, sd(tte_slow, na.rm = TRUE)]
sd_high <- pilot_cc[, sd(tte_high, na.rm = TRUE)]

# If SD is 0 or NA, use pooled SD or a small default
pooled_sd <- pilot_cc[, sqrt((var(tte_slow, na.rm = TRUE) + var(tte_high, na.rm = TRUE)) / 2)]
if (is.na(sd_slow) || sd_slow <= 0) sd_slow <- max(pooled_sd, 1)
```

```

if (is.na(sd_high) || sd_high <= 0) sd_high <- max(pooled_sd, 1)

effect_pilot <- mean_high - mean_slow

# Within-subjects: difference and SD of difference
pilot_cc[, diff_tte := tte_high - tte_slow]
mean_diff_pilot <- pilot_cc[, mean(diff_tte)]
sd_diff_pilot <- pilot_cc[, sd(diff_tte)]
if (is.na(sd_diff_pilot) || sd_diff_pilot <= 0) sd_diff_pilot <- max(pooled_sd, 1)

pilot_summary <- data.table(
  metric = c("Slow mean", "High mean", "Effect (high - slow)", "SD slow", "SD high", "Mean diff", "SD diff"),
  value = c(mean_slow, mean_high, effect_pilot, sd_slow, sd_high, mean_diff_pilot, sd_diff_pilot)
)

kable(pilot_summary, digits = 2, caption = "Pilot-based estimates for plank time (seconds)")

```

Table 1: Pilot-based estimates for plank time (seconds)

metric	value
Slow mean	106.20
High mean	121.80
Effect (high - slow)	15.60
SD slow	33.57
SD high	54.32
Mean diff	15.60
SD diff	27.65

3. Simulation Functions

We now define functions to simulate between-subjects and within-subjects designs using the pilot-based parameters.

```

SAMPLE_SIZES <- seq(20, 300, by = 20)
N_SIM <- 1000
ALPHA <- 0.05

# Between-subjects: one observation per person (slow vs high as two groups)
run_sim_between <- function(N, mean_slow, mean_high, sd_slow, sd_high) {
  n_per_group <- floor(N / 2)
  if (n_per_group < 1) return(NA_real_)

  y_slow <- rnorm(n_per_group, mean = mean_slow, sd = sd_slow)
  y_high <- rnorm(n_per_group, mean = mean_high, sd = sd_high)

  d <- data.table(
    y = c(y_slow, y_high),
    treat = rep(0:1, each = n_per_group)
  )

  mod <- lm(y ~ treat, data = d)
  ci <- coefci(mod, vcov = vcovHC(mod, type = "HC3"), level = 1 - ALPHA)
  lb <- ci["treat", 1]
}

```

```

ub <- ci["treat", 2]

as.numeric(!(lb <= 0 && ub >= 0))
}

# Within-subjects: paired slow vs high per person
run_sim_within <- function(N, mean_diff, sd_diff) {
  diff_sim <- rnorm(N, mean = mean_diff, sd = sd_diff)
  test_result <- t.test(diff_sim, mu = 0)
  as.numeric(test_result$p.value < ALPHA)
}

```

4. Power Estimation from Pilot

We estimate power across a range of sample sizes for both designs.

```

set.seed(241)

power_between <- sapply(SAMPLE_SIZES, function(n) {
  mean(replicate(N_SIM, run_sim_between(n, mean_slow, mean_high, sd_slow, sd_high)))
})

power_within <- sapply(SAMPLE_SIZES, function(n) {
  mean(replicate(N_SIM, run_sim_within(n, mean_diff_pilot, sd_diff_pilot)))
})

power_results <- data.table(
  N = rep(SAMPLE_SIZES, 2),
  Power = c(power_between, power_within),
  Design = rep(c("Between-subjects", "Within-subjects"), each = length(SAMPLE_SIZES))
)

power_wide <- dcast(power_results, N ~ Design, value.var = "Power")
kable(power_wide, digits = 3, caption = "Estimated power by sample size and design (pilot-based)")

```

Table 2: Estimated power by sample size and design (pilot-based)

N	Between-subjects	Within-subjects
20	0.100	0.675
40	0.174	0.932
60	0.238	0.985
80	0.320	0.998
100	0.412	1.000
120	0.467	0.999
140	0.510	1.000
160	0.581	1.000
180	0.621	1.000
200	0.682	1.000
220	0.722	1.000
240	0.771	1.000
260	0.786	1.000
280	0.826	1.000
300	0.828	1.000

5. Power Curves

Finally, we visualize the power curves for between-subjects and within-subjects designs based on the pilot estimates.

```
ggplot(power_results) +  
  aes(x = N, y = Power, color = Design) +  
  geom_line(size = 1) +  
  geom_point() +  
  geom_hline(yintercept = 0.8, linetype = "dashed", color = "red") +  
  scale_y_continuous(limits = c(0, 1)) +  
  labs(  
    title = "Pilot-Based Power Curves: Slow vs High-Tempo Music",  
    x = "Sample size (N)",  
    y = "Estimated power"  
  ) +  
  theme_minimal()
```

